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Category-Level Adversaries for Semantic Domain Adaptation

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ABSTRACT Recent advances in deep learning, especially deep convolutional neural networks, have led to great performance improvement over semantic segmentation systems. Unfortunately, training deep neural networks (DNNs) requires a humongous amount of labeled data, which is laborious and costly to collect and annotate. Thus, plenty of works have proposed an alternative solution to ease the training set creation by using synthetic data. However, models trained on these kinds of data usually under-perform on real images due to the well-known issue of domain shift. To address it, we propose a generative adversarial network (GAN)-based framework called category-level adversarial adaptation networks (CAA-Nets) for domain adaptation in the context of semantic segmentation. Considering semantic predictions that contain spatial and structure information of images, our idea is to make use of this character by imposing discriminators on the semantic predictions. Different from existing works, the proposed framework utilizes a category-level discriminator in the output space to shrink the gap between real and synthetic images. Similar to reinforcement learning, we take final results as a guide to update parameters in the right direction by use of the output-based discriminator. Moreover, to further enhance the performance, we construct an image-based generator and discriminator pair to distill the feature representations obtained by a DNN. Taking advantage of these modules, our model can achieve competitive performance compared with some existing methods. To showcase the generality and scalability of our approach, we evaluate the proposed method on the Cityscapes dataset by adapting from GTAV and SYNTHIA datasets, where the results demonstrate the effectiveness of our method.

INDEX TERMS Generative adversarial networks, domain adaptation, semantic segmentation.

I. INTRODUCTION

Deep neural networks (DNNs) have widely been applied in the field of computer vision and achieved the best performance in many tasks such as image classification [31], semantic segmentation [2], etc. Along with network layers increasing, the performance of models has a rapid boost, such as Resnet [31] and Densenet [16]. Nevertheless, the achievement builds on the basis of large paired datasets, which cost expensive human labors [1], [2] especially for the task of semantic annotations. Semantic segmentation aims to assign each pixel of a photograph to a semantic class

label while annotated labels need a large amount of dense pixel-level annotations obtained by experts. For instance, each Cityscapes [29] image on average takes about 90 minutes to annotate, highlighting the level of difficulty.

An alternative solution to solve the issue is to seek an automatic data generation approach. Rather than relying on expensive labors on annotating real-world data, recent progresses in Computer Graphics [3]–[5] make it possible to automatically capture both images and their corresponding semantic labeling from video games, e.g., Grand Theft Auto V (GTAV), which is a realistic open-world game based on Los Angeles. It provides an easy way to collect diverse labeled data whose quantity is unlimited. Therefore, it is very appealing to explore a technology, i.e., using synthetic data for

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neural networks training. In practice, utilizing synthetic data is indeed an effective method to address the issue of collecting a large amount of labeled data that needs expensive human labors. However, models trained on synthetic data fail to perform well on real-world datasets owing to the domain gap between datasets. Domain adaptation focuses on minimizing the domain gap and avoids the degeneration of performance at inference time where no labels from target domain are available. Hence, this paper is dedicated to developing domain adaptation algorithms for the task of semantic segmentation, which is commonly referred to as Unsupervised Domain Adaptation (UDA).

Traditional approaches for domain adaptation, i.e., Maximum Mean Discrepancy (MMD) [6], [7], involve minimizing some measure of distance between the source and the target distributions. More recent researches [8]–[10] focus on integrating Generative Adversarial Networks (GANs) [11], [12], [18], [33] to minimize the domain gap on feature distributions so that both data distributions are undistinguished. These methods have achieved great improvement in the classification problems and have a common similarity of utilizing adversarial learning at feature-level representations or pixel-level images. Actually, adversarial learning on feature space is not strong enough to guarantee that the feature of target domain learned by networks is correctly mapped. Different from source domain, which has paired images and labels, target domain can not force networks to extract correctly mapped feature representations through supervised learning. Therefore, existing models have the misclassification problem and result in the degeneration of performance when applied to target domain.

To mitigate the issue above, we consider the task of semantic segmentation from another aspect. Similar to reinforcement learning, our model takes final results as a guider for segmentation networks. Although source and target images appear difference in image styles, their semantic labels are similar to each other. Considering that semantic segmentation aims to get high-accurate dense labels, so why do not directly integrate GANs in the output space of segmentation networks. Unlike prior works that apply adversarial learning on feature-level representations or pixel-level images, our idea is to apply adversarial learning on final semantic predictions. We expect that the model can take advantage of GANs to make semantic predictions of both domains close enough so that the prediction of target domain can fool discriminators as source domain. Different from Tsai *et al.* [25], the proposed model takes a slice of the prediction as the input of a discriminator, i.e., one class map instead of all class maps. Thus it is named Category-level Adversarial Adaptation Networks (CAA-Net). Specifically, we regard the discriminator as a multi-category classifier and expect the discriminator to distinguish the slice of source predictions from that of target predictions. The classifier can learn the prior knowledge of every class (e.g., cars) in source domain with labeled data. According to the structure and location information of a class map, it has the ability to distinguish

which class the map belongs to. To avoid the occurrence of incorrect mappings in feature space, our model utilizes the classifier as a tutor to guide target semantic predictions when processing target data. In a word, the formulation is to utilize the classifier under adversarial supervision so as to make parameters of segmentation networks update in the right direction. In addition, inspired by [32], we employ another two discriminators as style classifiers and semantic discriminators respectively. Style classifiers aim to distinguish real images from generated images while semantic discriminators act as minor segmentation networks. Both discriminators are crucial to obtain domain-invariant feature representations.

The main contribution of this work is that we propose a novel GAN-based framework that combines the backbone with GANs to apply adversarial learning in the output space of segmentation networks. Unlike most of previous works, it does not impose discriminators on feature space for global alignment. Instead, our model employs a multi-category discriminator on the output to distinguish whether semantic predictions are from source domain. Using this idea, the proposed method imposes more effective constraints on semantic predictions to mitigate the issue of misclassification. Second, we use the self-reconstruction strategy for both domains by employing a generator to reconstruct source and target images from intermediate feature representations obtained by a DNN. Combining with another two discriminators based on images, the backbone has the ability to extract semantic information with no specific-domain components. Third, extensive experiments are conducted to show that our model is able to achieve competitive performance compared with some existing methods in terms of accuracy. Furthermore, our model does not introduce extra computational costs at inference time because it only needs the backbone to predict labeling results.

II. RELATED WORKS

A. SEMANTIC SEGMENTATION

Semantic segmentation is a task of assigning a semantic class label to each pixel in an image. Traditional methods rely on local image features handcrafted by domain experts [13]. Until the seminal networks [2] proposed by Long *et al.*, the fully convolutional network achieves great performance improvement and is widely implemented in the field of semantic segmentation. Most recent top-performing methods follow upon FCN [1], [14]. In recent years, we have witnessed a large amount of great works that have an obvious boost on the performance of semantic segmentation. Dilated convolution [17] is proposed to resolve the contradictory between larger receptive fields and larger feature map resolution. DeepLab [1] incorporates Conditional random field (CRF) with convolutional neural networks (CNN) to reason about spatial relationship. Recently, the state-of-the-art work [15] has achieved a remarkable result by combining dilated convolution with DenseNet [16].

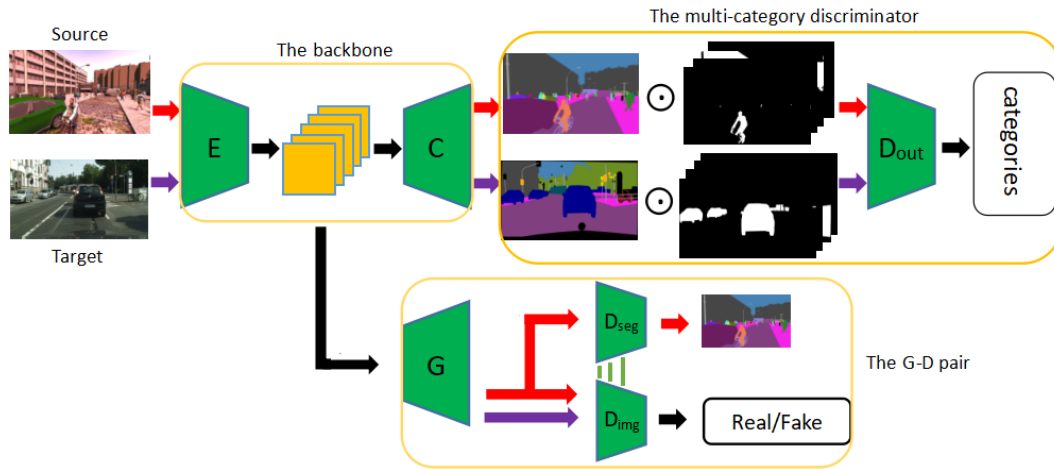


FIGURE 1. The flowchart of our category-level adversarial adaptation network (CAA-Net), including three modules: The backbone, the multi-category discriminator and the G-D pair. Red arrow lines represent the information flow of source images and purple ones for target images. Black arrow lines represent the common information flow of both domains. Green blocks respectively represent encoder E , classifier C , generator G and discriminator D_{out} , D_{img} , D_{seg} . Among these blocks, D_{img} and D_{seg} share the parameters of low-level layers. To make target predictions closer to the source ones, we utilize a multi-category discriminator D_{out} to distinguish whether the input is from the source or target domain. D_{out} takes semantic predictions of net C multiplied by corresponding masks as input and regularizes the learned model by using a cross-entropy loss. The $\otimes$ represents a mathematical operator of element-wise multiplication.

B. DOMAIN ADAPTATION

Conventional machine learning has a basic assumption that training and test data are sampled independently from an identical distribution. That is to say that training and test data come from the same dataset. However, this assumption is not always reasonable in real world scenarios because of the complexity and diversity of real scenarios. Methods of domain adaptation for image classification have been developed to solve the problem of domain shift that often leads to a significant performance drop at inference time. In computer vision, numerous methods [19]–[21] have been proposed to mitigate the impact of domain gap and these approaches conduct the same operation, i.e., aligning the feature distributions between source and target images. Recently, the PixelDA method [22] addresses domain adaptation for image classification by transferring source images to target domain, thereby creating a simulated dataset for supervised learning. In conclusion, there are two major methods applied to domain adaptation, one aiming to align the feature distributions between domains, the other aiming to create a simulated dataset for target domain by generating target-like images from source domain.

C. DOMAIN ADAPTATION FOR SEMANTIC SEGMENTATION

So far most works in domain adaptation focus on the task of image classification. Not until recently has the community paid attention to domain shift problem in semantic segmentation, which has a significant impact on auto-driving or robotic systems. Few works have explicitly studied domain adaptation for semantic segmentation. Hoffman *et al.* [23] implement adversarial learning in a fully convolutional way on feature representations to open a good start for the task. Zhang *et al.* [30] propose curriculum-style learning where

a teacher network solve the easier task of learning global label distributions over images and local distributions over landmark superpixels, then a student segmentation network is trained so that the target label distribution follow these inferred label properties. Similar to PixelDA [22], CyCADA [24] creates a simulated training set for target domain by transforming real source images to fake target images. Reference [36] takes the advantage of additional geometric information from synthetic data, a powerful yet largely neglected cue, to bridge the domain gap. Reference [34] introduces a category-level adversarial network, aiming to enforce local semantic consistency during the trend of global alignment. Reference [35] proposes a novel bidirectional learning framework for domain adaptation of segmentation. Using the bidirectional learning, the image translation model and the segmentation adaptation model can be learned alternatively and promote to each other. Besides, Tsai *et al.* [25] construct a multi-level adversarial network to effectively perform output space domain adaptation at different feature levels. Note that our proposed method is different from that proposed by Tsai *et al.* because ours is based on category-level adversarial learning where it takes a slice of the prediction as input, rather than the entire prediction.

III. METHOD

A. OVERVIEW OF THE PROPOSED MODEL

Our method for domain adaptation focuses on aligning the feature distributions between domains by integrating adversarial learning in the output space. We expect that the backbone can extract distilled semantic information and acquire higher accurate labels by virtue of the power of adversarial networks. As shown in Fig. 1, our model consists of three branches: the backbone, the image-based G-D pair and the

multi-category discriminator. The backbone is similar to a pre-trained model such as VGG-16. It consists of two parts: a feature extractor E and a classifier C . The image-based G-D pair is composed of one generator for reconstructing RGB images from learned feature representations and two discriminators: D_{img} used to distinguish real images from reconstructed images, D_{seg} used to maintain semantic information. The output-based multi-category discriminator D_{out} is a classifier whose input is a slice of the prediction, i.e., one channel of the prediction. It achieves that the data distribution of target predictions is as close as possible to that of source predictions.

B. THE FLOW OF SOURCE AND TARGET DATA

Consider source images x^s and corresponding labels y^s . Note that y^s is a pixel-level label that contains class information in the case of semantic segmentation. For the backbone, the classifier C takes the feature representations $E(x^s)$ as input and produces image-sized label predictions $\hat{y}^s$, which are under supervision by using a cross-entropy loss with real labels y^s . Meanwhile the generator G reconstructs source images $G(E(x^s))$ from the feature representations $E(x^s)$. The discriminator D_{img} is trained to distinguish real source images x^s from generated source images $G(E(x^s))$. It plays a competitive game with G to make generated images more realistic. To maintain intact semantic information, $G(E(x^s))$ are also under supervision by D_{seg} to avoid the problem where classes in $G(E(x^s))$ are incorrectly mapped to other classes. Besides, our output-based multi-category discriminator D_{out} takes every masked channel of semantic predictions $C(E(x^s))$ as input to classify its category. For example, $C(E(x^s))$ is a 4-D tensor in Pytorch and we regard $C(E(x^s))[:, 0, :, :]$, $C(E(x^s))[:, 1, :, :]$, $C(E(x^s))[:, 2, :, :]$ as class 0, 1, 2 respectively. Also, it is similar to other channels of predictions $C(E(x^s))$.

Consider target images x^t that have no corresponding annotations. Similar to source images, target images x^t are fed to net E to acquire feature representations $E(x^t)$ and the classifier C takes $E(x^t)$ as input to output image-sized label predictions $\hat{y}^t$, which are not under supervision because of no labels available. Meanwhile G reconstructs target images $G(E(x^t))$ from the feature representations $E(x^t)$. Note that we use one image-based discriminator to classify the style of images for both domains. Thus D_{img} is also trained to distinguish real target images x^t from generated target images $G(E(x^t))$. Also, D_{out} takes every masked channel of predictions $C(E(x^t))$ as input and gives a feedback loss to the backbone so as to have a good adaptation for target domain.

C. THE BACKBONE

In this work, we use FCN-8s [2] as the backbone. To enlarge the receptive field, the model removes pooling layers and applies dilated convolution layers [17] in conv4 and conv5 layers with a stride of 2. The weights of the backbone are initialized with the VGG-16 [26] model trained on Imagenet [27]. The backbone is split into two parts: the front

part of the backbone is regarded as a feature extractor denoted by E and the rest part is a pixel-wise classifier denoted by C whose input is the output of E . The output of C is a label map up-sampled to the same size as input images. For the task of semantic segmentation, the key module is the backbone which is utilized to produce final results at inference time. In addition, the core technique of semantic domain adaption is to induce E to acquire distilled feature representations $E(x)$ and improve the classification performance of C . For source data, we use paired data to learn the parameters of the backbone through supervised learning. On the other hand, we freeze the parameters of G and force reconstructed source images $G(E(x^s))$ to fool D_{img} as target images for obtaining domain-invariant feature representations. Meanwhile, the minor segmentation network D_{seg} is introduced to keep the intact semantic information. Hence, the objective l_{src} about source data is expressed as:

$$l_{src}(E, C) = H(C(E(x^s)), y^s) + \lambda_{seg}H(D_{seg}(G(E(x^s))), y^s) + \lambda_{adv}H(D_{img}(G(E(x^s))), Y_{real-tgt}) \quad (1)$$

where $H(p, q)$ is a pixel-wise cross-entropy loss function and Y is a style label which has the same size as the output of D_{img} . In this work, we set the style label $Y_{real-src}$ zero for real source images and it is similar to others, i.e., one for real target images, two for reconstructed source images, three for reconstructed target images. The hyper-parameters λ_{seg} , λ_{adv} control the weights of different objective terms.

For target data, we force reconstructed target images $G(E(x^t))$ to fool D_{img} as source images. This operation further distills the learned feature representations $E(x)$ of both domains with no domain-specific components. Besides, the output-based multi-category discriminator D_{out} is directly imposed on semantic predictions to achieve that the data distribution of target predictions is undistinguished from the source one. The objective l_{tgt} about target data is expressed as:

$$l_{tgt}(E, C) = \lambda_{adv}H(D_{img}(G(E(x^t))), Y_{real-src}) + \lambda_{adv} \sum_{n=1}^C H(D_{out}(M_n^t), Label^n) \quad (2)$$

where C represents the total number of categories (i.e., 19 for Cityscapes dataset). $Label^n$ is the label of corresponding masked feature map M_n^t or M_n^s and its label number is $n - 1$.

$$M_n^t = C_n(E(x^t)) \odot mask_n^t \quad (3)$$

M_n^t derives from element-wise multiply between the feature map $C_n(E(x^t))$ and the mask $mask_n^t$. Specifically, $C_n(E(x^t))$ is the n -th feature map of $C(E(x^t))$, i.e., $C(E(x^t))[:, n - 1, :, :]$. $mask_n^t$ has the same size as $C_n(E(x^t))$ and its elements are either one or zero. For $mask_n^s$, according to real source labels y^s , we assign one to those pixels belonging to the n -th class, and zero to other pixels. For $mask_n^t$, our model uses the pseudo label of $C(E(x^t))$ as a guider to acquire corresponding masks

because of no real labels available for target domain. Note that M_n^t is for target domain and M_n^s is for source domain.

According to the flow of source and target data, the overall loss function of the backbone is expressed as:

$$L_{E,C} = l_{src} + l_{tgt} \quad (4)$$

D. THE IMAGE-BASED GENERATOR AND DISCRIMINATOR PAIR

Our model constructs the image-based generator and discriminator pair for two goals: 1) It expects the learned feature representations $E(x^s)$ and $E(x^t)$ to maintain intact semantic information by reconstructing images $G(E(x^s))$ and $G(E(x^t))$. 2) Another goal is to acquire the distilled feature representations $E(x^s)$ and $E(x^t)$ that have no domain-specific information. In this work, G is designated to reconstruct images $G(E(x^s))$ and $G(E(x^t))$ for both domains instead of using two domain-specific generators, one for source domain, the other for target domain. D_{img} is utilized to distinguish reconstructed images $G(E(x^s))$ and $G(E(x^t))$ from real images x^s and x^t . The objective of D_{img} is expressed as:

$$\begin{aligned} L_{D_{img}}(D_{img}) = & H(D_{img}(x^s), Y_{real-src}) \\ & + H(D_{img}(G(E(x^s))), Y_{fake-src}) \\ & + H(D_{img}(x^t), Y_{real-tgt}) \\ & + H(D_{img}(G(E(x^t))), Y_{fake-tgt}) \end{aligned} \quad (5)$$

Besides, a minor segmentation net D_{seg} , which has partly-shared weights with D_{img} , is introduced to make ensure that the distilled feature representations still maintain intact semantic information. Meanwhile, D_{seg} facilitates the G-D pair to realize a more stable GAN training. The objective of D_{seg} is expressed as:

$$L_{D_{seg}}(D_{seg}) = H(D_{seg}(x^s), y^s) \quad (6)$$

On the other hand, G is compelled to identify domain-specific information from feature representations $E(x)$ and reconstruct corresponding images $G(E(x))$ by playing a minmax game with D_{img} . This process can guarantee that G has a nice identification for both domains. That is to say if the learned feature representations include domain-specific information, G will find domain information and reconstruct images of corresponding domains. Depending on the nice performance of G-D pair, it is convenient for net E to acquire distilled feature representations when updating the backbone. From another aspect, the G-D pair could be regarded as an auxiliary network. Unlike most works that impose adversarial networks on the feature representations, we use D_{img} and D_{seg} to facilitate our model to acquire undistinguished feature representations. In summary, the objective of G consists of three parts: an adversarial loss $l_{adv,G}$, a segmentation loss $l_{seg,G}$ and a L1 reconstruction loss $l_{rec,G}$.

$$\begin{aligned} l_{adv,G} = & H(D_{img}(G(E(x^s))), Y_{real-src}) \\ & + H(D_{img}(G(E(x^t))), Y_{real-tgt}) \end{aligned} \quad (7)$$

$$l_{seg,G} = H(D_{seg}(G(E(x^s))), y^s) \quad (8)$$

$$l_{rec,G} = \|x^s - G(E(x^s))\|_1 + \|x^t - G(E(x^t))\|_1 \quad (9)$$

The overall loss function of G is expressed as:

$$L_G(G) = l_{rec,G} + l_{seg,G} + \lambda_{adv} l_{adv,G} \quad (10)$$

where λ_{adv} is the weight used to balance $l_{adv,G}$ with $l_{rec,G}$ and $l_{seg,G}$.

E. THE OUTPUT-BASED MULTI-CATEGORY DISCRIMINATOR

The motivation to design the output-based multi-category discriminator is that we consider the semantic prediction is composed of pixel-level feature maps, which consist of 19 channels for Cityscapes dataset. Every channel is corresponding to a feature map of one class. For example, the prediction $C(E(x))$ is a 4-D tensor and $C(E(x))[:, i, :, :]$ ($0 \leq i \leq 18$) is a pixel-level feature map of a single category (cars, sky, buildings or other categories). It is assumed that these feature maps contain the structure and location information of objects, which have the same semantic information as real labels. Furthermore, with the rapid development of deep learning, we have realized that generative adversarial networks (GANs) have a good application on images processing. The discriminator tries to distinguish real images from generated images and the corresponding generator utilizes an adversarial loss to force generated images to be undistinguished from the ground truth. Based on the analysis above, GANs are utilized to make the data distribution of target predictions $C(E(x^t))$ close to that of source predictions $C(E(x^s))$. Here, due to the limited memory, we use a minor ResNet as the discriminator. In this work, predictions of source images and target images are treated as real samples and fake samples respectively by the output-based multi-category discriminator D_{out} . Unlike traditional discriminators that output zero or one, the proposed discriminator is utilized as a multi-category classifier using a cross-entropy loss for both domains (i.e., source and target). The loss can be written as:

$$\begin{aligned} L_{D_{out}}(D_{out}) = & \sum_{n=1}^C H(D_{out}(M_n^s), Label^n) \\ & + \lambda_{out} \sum_{n=1}^C H(D_{out}(M_n^t), Label^{tgt}) \end{aligned} \quad (11)$$

$$M_n^s = C_n(E(x^s)) \odot mask_n^s \quad (12)$$

$$M_n^t = C_n(E(x^t)) \odot mask_n^t \quad (13)$$

where $Label^{tgt}$ has the same size as $Label^n$ but different labels, i.e., $Label^{C+1}$. The hyper-parameter λ_{out} is the weight used to balance the loss between source and target data. Our goal is to obtain high-accurate dense labels, so the proposed loss $L_{D_{out}}$ takes final results as a guider for segmentation networks. Unlike prior works that apply adversarial learning on feature-level representations or pixel-level images, a cross-entropy loss is introduced in the output space of segmentation network to guarantee that the feature of target domain learned

Algorithm 1 CAA-Net Training Procedure

Input: Source data S , target data T , K (number of iterations).
Initialize the parameters of CAA-Net.
for $c=0$ to $K-1$ **do**
 sample source images x^s , source labels y^s from S
 and target images x^t from T .
 update parameters of D_{img} to minimize Equation 5.
 update parameters of D_{seg} to minimize Equation 6.
 update parameters of D_{out} to minimize Equation 11.
 update parameters of G to minimize Equation 10.
 update parameters of E , C to minimize Equation 4.
end
Output: Trained CAA-Net.

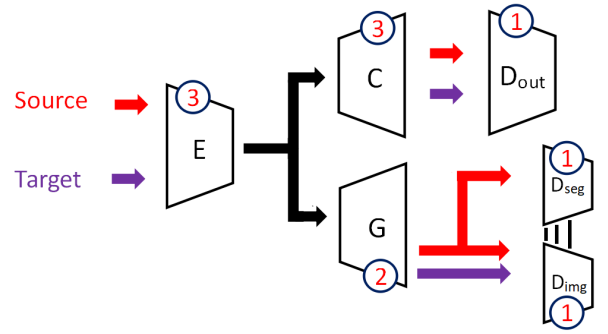


FIGURE 2. The direction of data flow during the forward pass. Every block in the network is presented with a number, such as one, two, three, which indicates the update order of the block. Red arrow lines represent the information flow of source images and purple ones for target images. Black arrow lines represent the common information flow of both domains.

by networks is correctly mapped. In a word, the formulation is to utilize the classifier under adversarial supervision so as to make parameters of segmentation networks update in the right direction. This strategy has a nice effect on mitigating the misclassification problem.

F. NETWORKS TRAINING

To train the proposed adaptation model, we use a multi-step training strategy of K iterations, as shown in Alg. 1. In each training batch, D_{img} , D_{seg} and D_{out} are first trained. D_{img} uses a combination of several adversarial terms $L_{D_{img}}$ to update parameters for the goal of differentiating real images from generated images. Meanwhile, a cross-entropy loss $L_{D_{seg}}$ over source domain is applied to update parameters for D_{seg} by supervised learning. D_{out} also utilizes the adversarial mechanism to make the data distribution of target predictions $C(E(x^t))$ close to that of source predictions $C(E(x^s))$ by use of a multi-category cross-entropy loss denoted by $L_{D_{out}}$. Next step, we focus on the parameters learning of G that has a significant impact on the whole model. In this step, G utilizes a combination of an adversarial loss $l_{adv,G}$, a reconstruction loss $l_{rec,G}$ and a segmentation loss $l_{seg,G}$ to update parameters. The overall loss L_G imposes a strong constraint on G to have a nice render effect on generated images. Basing on the modules described above, we finally update the parameters of the backbone, using a loss denoted by $L_{E,C}$. To make it easier to understand the algorithm, please see Fig. 2. In each iteration, it has three update steps following the order from one to three. During the update step, only parameters of corresponding block are updated and other blocks are frozen. For example, in the first step, we update the parameters of D_{img} , D_{seg} and D_{out} while keeping other parameters unchanged.

IV. EXPERIMENTS

In this section, we present the experimental setup, implementation details and compare the results of our approach with some existing methods. Ablation study is conducted to further prove the effectiveness of the proposed method.

A. DATASETS

In our experiments, two challenging synthetic datasets: GTAV [3] and SYNTHIA [28] are considered. We use Cityscapes dataset [29] as target domain, SYNTHIA or GTAV as source domain. Following prior works [23], [30], we use 500 images from the Cityscapes val set to evaluate our model. All of datasets are publicly available.

Cityscapes dataset is a challenging real-world dataset. It contains urban street images collected from a moving vehicle captured in 50 cities around Germany and neighboring countries. The dataset comes with 5,000 annotated images split into three sets, i.e., 2,975 images in the training set, 500 images in the val set and 1,595 images in the test set.

GTAV is a dataset recently proposed for learning from synthetic data. It has 24,966 photo-realistic images rendered by the gaming engine Grand Theft Auto V (GTAV). The ground truth annotations are compatible with Cityscapes dataset.

SYNTHIA is a large dataset of photo-realistic frames rendered from a virtual city with precise pixel-level semantic annotations. In this work, we use the SYNTHIA-RAND-CITYSCAPES subset that contains 9,400 images with annotations that are compatible with Cityscapes dataset.

Experiment Setup: For the task of GTAV $\rightarrow$ CITYSCAPES, we train our model on GTAV and validate it on Cityscapes dataset. Specifically, it uses annotated images as source domain and unlabeled Cityscapes train set as target domain. We consider the 19 common classes shared with Cityscapes dataset: bike, fence, wall, traffic sign, pole, motorcycle, traffic light, sky, bus, rider, vegetation, terrain, train, building, car, person, truck, sidewalk and road.

For the task of SYNTHIA $\rightarrow$ CITYSCAPES, similar to the GTAV $\rightarrow$ CITYSCAPES setting, our model is trained on SYNTHIA and validated on Cityscapes dataset. We consider the 16 common classes shared with Cityscapes dataset: sky, building, road, sidewalk, fence, vegetation, pole, car, traffic sign, person, bicycle, motorcycle, traffic light, bus, wall, and rider.

TABLE 1. Comparison results for the semantic segmentation of the Cityscapes images by adapting from GTAV. The results are reported over the 19 common classes. The underlined numbers indicate the highest mIoU values among all of methods.

method	road	sidewalk	building	wall	fence	pole	t light	t sign	veg	terrain	sky	person	rider	car	truck	bus	train	mbike	bike	mIOU
NoA	31.9	18.9	47.7	7.4	3.1	16.0	10.4	1.0	76.5	13.0	58.9	36.0	1.0	67.1	9.5	3.7	0.0	0.0	0.0	21.1
Wild	70.4	<u>32.4</u>	62.1	14.9	5.4	10.9	14.2	2.7	<u>79.2</u>	21.3	64.6	<u>44.1</u>	4.2	70.4	8.0	7.3	0.0	3.5	0.0	27.1
CDA	74.9	22.0	71.7	6.0	11.9	8.4	<u>16.3</u>	<u>11.1</u>	75.7	13.3	66.5	38.0	<u>9.3</u>	55.2	18.8	<u>18.9</u>	0.0	<u>16.8</u>	<u>16.6</u>	28.9
CYC	78.9	28.3	72.0	27.8	11.5	<u>18.5</u>	11.6	10.8	75.1	30.1	63.2	35.3	1.7	72.4	11.8	5.1	3.7	4.9	0.6	29.6
LSD	79.6	25.7	73.8	30.2	13.1	16.9	11.7	5.5	74.5	28.6	68.2	36.7	3.9	74.1	12.3	2.74	<u>8.27</u>	9.0	1.1	30.3
Ours	<u>80.9</u>	23.7	<u>76.3</u>	<u>30.4</u>	<u>15.6</u>	15.8	12.0	3.9	76.0	<u>32.7</u>	<u>71.2</u>	33.8	2.3	<u>74.5</u>	<u>20.6</u>	9.1	2.2	6.2	0.3	<u>30.9</u>

B. IMPLEMENTATION DETAILS

We implement our network using the Pytorch toolbox on a single 1080Ti GPU with 11 GB memory. Our model is trained for 100,000 iterations using Adam solver with a batch size of 1. Learning rate of 2×10^{-5} is used for E and C networks, and 10^{-4} for G and D networks. Due to the GPU memory limitation, the images used in our experiments are resized to 128×64 , 256×128 , 512×256 respectively. While evaluating on Cityscapes dataset whose images and ground truth annotations are of size 2048×1024 , we first produce our predicted labels with the same size as input images and then upsample them to final labels that are of size 2048×1024 for evaluation. In our experiments, the hyper-parameters: λ_{adv} , λ_{seg} , λ_{out} are set to 0.1, 0.1 and 0.05, respectively.

C. BASELINES

In our experiments, we compare our approach to some competing methods that share the training code of adapted segmentation networks based on VGG-16 for fair comparison.

No adaptation (NoA) [23]. We directly train the FCN-8s model on SYNTHIA and GTAV without any domain adaptation. This is the most basic baseline for our experiments.

FCNs in the wild (Wild) [23] introduces a pixel-level adversarial loss to the intermediate layers of the network and impose constraints on label statistics to the network output. It use the front-end dilated fully convolutional network based on VGG-16 as the baseline model.

Curriculum Domain Adaptation (CDA) [30] proposes a curriculum-style learning approach to minimize the domain gap in semantic segmentation. The curriculum domain adaptation first solves easy tasks such as estimating label distributions, then infers the necessary properties about the target domain.

Cycle-Consistent Adversarial Domain Adaptation (CYC) [24] creates a simulated training set for target domain by transforming real source images to fake target images. It adapts representations at both the pixel-level and feature-level. Here we use the transformed GTAV dataset to train the model at size 512×256 for the task of GTAV $\rightarrow$ CITYSCAPES.

AdaptSegNet [25] adopts adversarial learning in the output space. To further enhance the adapted model, they construct a multi-level adversarial network to effectively perform output space domain adaptation at different feature levels.

Learning from Synthetic Data (LSD) [9] first projects the intermediate feature representations obtained by a CNN to the image space and then imposes the domain alignment constraint on networks to learn domain-invariant features, which is accomplished by employing a series of adversarial losses.

D. GTAV $\rightarrow$ CITYSCAPES

In this experiment, GTAV dataset and Cityscapes dataset are set as source domain and target domain respectively. We randomly pick 1,000 images from 24,966 labeled images for validation and use the rest of images for training. Meanwhile, unlabeled images of the Cityscapes training set are used to adapt our model to target domain. In order to ensure the fairness of experimental results, we follow the evaluation protocol as specified by previous works [23], [30]. 19 common classes between GTAV and Cityscapes dataset are adopted as our labels. Table 1 shows evaluation results of the proposed algorithm against the baseline methods. Our model yields a mIoU of 30.9% and outperforms the baselines. Also, it can be observed that our model achieves the highest performance in most categories. Only some classes (e.g., t light, t sign) are not often seen during training time such that our model can not learn proper knowledge from the limited information. The experimental results suggest that the proposed model can produce competitive results and perform even better compared with baselines. In addition, we present comparisons of mean IoU gap between adapted and unadapted results in Table 2. Conditioned on the same setting, our model not only achieves the highest mean IoU, but also the largest gap up to 8.4 points, suggesting the superiority of the proposed algorithm. Several representative results are shown in Fig. 4.

E. SYNTHIA $\rightarrow$ CITYSCAPES

In this experiment, we adopt SYNTHIA dataset as source domain, and Cityscapes dataset as target domain.

TABLE 2. Performance gap between adapted and unadapted results on the GTAV → CITYSCAPES setting. In addition to mean IoU values, we list the basic networks and image sizes used in comparative experiments. Note that both of them have a significant impact on the final performance.

Method	Baseline	Image size	Source only	Adapt	mIoU Gap
Wild	VGG-16	-	21.1	27.1	6.0
CDA	VGG-16	640x320	22.3	28.9	6.6
CYC	VGG-16	512x256	22.1	29.6	7.5
LSD	VGG-16	512x256	22.7	30.3	7.6
Ours	VGG-16	512x256	22.5	30.9	8.4

TABLE 3. Comparison results for the semantic segmentation of the cityscapes images by adapting from SYNTHIA. The results are reported over the 16 common classes. The underlined numbers indicate the highest mIoU values among all of methods.

method	road	sidewalk	building	wall	fence	pole	t light	t sign	veg	sky	person	rider	car	bus	mbike	bike	mIoU
NoA	6.4	17.7	29.7	1.2	0.0	15.1	0.0	7.2	30.3	66.8	51.1	1.5	47.3	3.9	0.1	0.0	17.4
Wild	11.5	19.6	30.8	<u>4.4</u>	0.0	<u>20.3</u>	0.1	11.7	42.3	68.7	<u>51.2</u>	3.8	<u>54.0</u>	3.2	0.2	0.6	20.2
CL	65.2	26.1	<u>74.9</u>	0.1	<u>0.5</u>	10.7	3.5	3.0	76.1	70.6	47.1	8.2	43.2	<u>20.7</u>	0.7	<u>13.1</u>	29.0
LSD	68.8	<u>29.7</u>	72.5	3.3	0.4	18.2	<u>7.7</u>	<u>11.9</u>	73.4	73.8	41.0	<u>14.2</u>	47.2	5.2	<u>8.4</u>	12.4	<u>30.5</u>
Ours	<u>70.8</u>	28.5	73.9	3.7	0.18	18.3	5.3	9.8	<u>77.1</u>	<u>77.5</u>	40.6	8.4	50.0	6.7	6.4	9.8	30.4

We randomly pick 100 images from 9,400 labeled images of SYNTHIA for validation and the rest of images are used for training. Different from GTAV, 16 common classes between SYNTHIA and Cityscapes dataset are treated as effective labels for training and validation. Table 3 presents some experimental results for adapted segmentation. We note that for small objects such as traffic lights and traffic signs, they are harder to adapt since they easily get merged with background classes. According to the data, it is easy to know that our model outperform baselines in some categories which are often seen in urban scenes and under-perform in some infrequent categories. We think it is attributed to two aspects: 1) Our model is prone to having a good adaptation for those frequent categories, such as road, vegetation, sky, etc. Nevertheless, for those infrequent categories such as motorbike, bike, our model is harder to learn the precise knowledge from the limited information so that our classifier has difficulty giving a correct judgment on semantic predictions. 2) It is well known that large datasets facilitate neural networks to get higher performance. Compared with GTAV dataset containing 24,966 images, SYNTHIA is a relatively smaller dataset, just a half of the former. Suffering from it, the performance of our model has a slight drop. In spite of all above, experimental results suggest that the proposed model can produce competitive results against baselines.

F. EFFECT OF IMAGE SIZE

The datasets used in this paper consist of images of large resolution which is atleast twice larger than the most commonly used segmentation benchmarks for CNNs, i.e., Pascal VOC(500 × 300) and MSCOCO (640 × 480). It is worth

saying that different images sizes have a significant impact on the performance of classification accuracy. Conditioned on the same algorithm and hyper-parameters setting, Fig.3(a) shows the classification performance of our model on multiple sizes. It is clear that performance makes great progress with larger images but at the price of computational cost and huge memory. Moreover, images of high resolution not only cost computation resource but also consume much more time. Hence, it heavily depends on the performance of hardware. In order to have an intuitive understanding to processing rate, Fig. 3(b) lists the consuming time (per image) on average at inference time. It is obvious that the expense of time does not have a linear relation with performance improvement. For instance, compared with images of 256×128 , images of 512×256 obtain a mean IoU of 30.9%, which has a 20% improvement. According to the trend of the performance curve in Fig. Fig. 3(a), we infer that every time the image size is doubled, the performance is increased by around 20%. Yet, its time curve has a different trend with the performance curve. Note that here we evaluate our model on images of 1024×512 just for more information about time costs. In light of the trend of the time curve, it is obvious that the evaluation time has an explosive rise with the resolution increasing. For images of 1024×512 , evaluation time (per image) is about 0.28s, which surpasses the time consumed by 512×256 images by around 65%. In view of the problem of limited processing rate and available memory in practice, we should have a trade-off between the performance and time costs. In this work, we use the maximal size 512×256 to get the best result. At inference time, it only needs the backbone to predict labeling results. The backbone has 134 million parameters

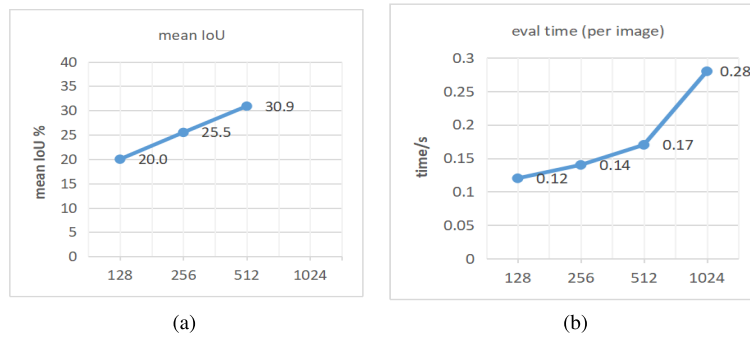


FIGURE 3. Study of the effect of image size on the GTAV → CITYSCAPES setting. (a) Mean IoU values across different image sizes. The dotted line represents the trend line. (b) Evaluation time across different image sizes.

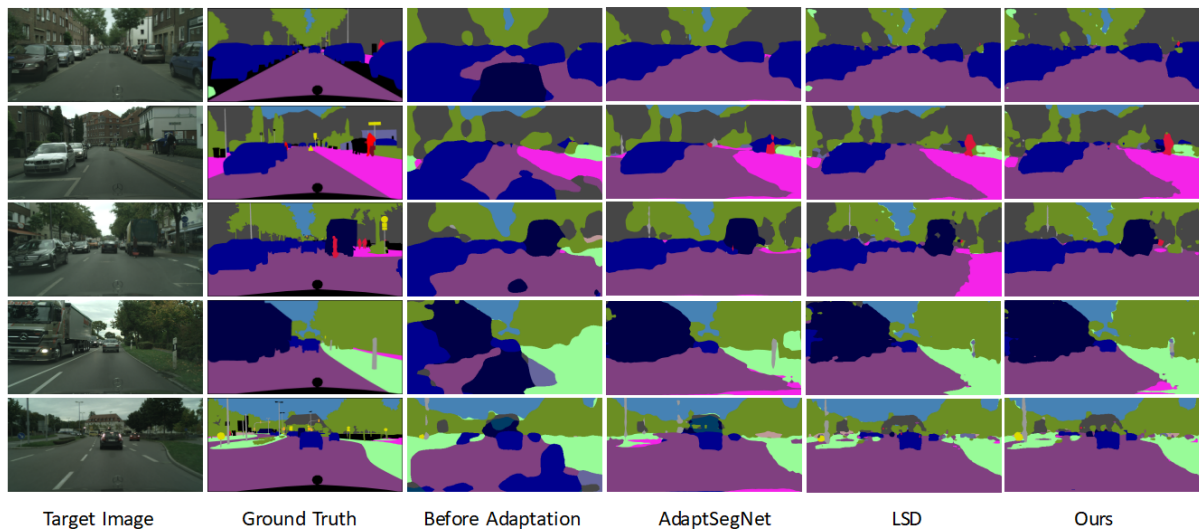


FIGURE 4. Example results of adapted segmentation for GTAV → Cityscapes.

in total, which is close to the number of parameters of VGG-16 network.

G. ABLATION STUDY

In this section, we carry out several experiments to show how each component in our loss function affects the final performance. Without changing hyper-parameters and the net structure, we only add or remove some component of the loss function for fair comparison. The following cases are considered: (a) Ours(full): the full implementation of our approach (b) Ours w/o D_{seg} : On the basis of Ours(full), the minor segmentation net D_{seg} is removed. (c) Ours w/o D_{out} : Here, we remove the loss introduced by the category-level discriminator D_{out} . (d) Ours w/o D_{img} : Instead of using the image-based discriminator D_{img} as a traditional discriminator that outputs a 2-D probability vector, i.e., zero or one, D_{img} is designated as a classifier where the output is a 4-D probability vector that is corresponding to four classes: source-real, target-real, source-fake and target-fake. In this setting, net D_{img} is removed and the reconstruction loss introduced

TABLE 4. Ablation study showing the effect of each component on the final performance of our approach on the GTAV → CITYSCAPES setting.

Method	mean IoU
Source-only	22.5
Ours w/o D_{img}	25.7
Ours w/o D_{out}	27.8
Ours w/o D_{seg}	29.3
Ours(full)	30.9

by G is reserved. (e) Source-only: we remove all of auxiliary networks and only employ the backbone. A multi-category cross-entropy loss for semantic segmentation is treated as the final loss function. The mean IoU results on the Cityscapes val set are shown in Table 4. According to the evaluation data, it can be observed that each component is very important to obtain the full improvement in performance.

Furthermore, we carry out additional comparison experiments on GTAV dataset to prove the effectiveness and generalization of the module D_{out} . Note that few works have

TABLE 5. Effects of the multi-category discriminator applied on other competitive models. The mean IoU values across different image size are evaluated on the GTAV → CITYSCAPES setting.

Image size	LSD	LSD+ D_{out}	mIoU gain
128x64	18.9	19.6	0.7
256x128	24.0	25.2	1.2

shared their codes in the community in the field of semantic domain adaptation. In this experiment, we choose the available LSD [9]

as the baseline and the model combined with D_{out} is referred to as $LSD + D_{out}$. Without changing nets architecture and hyper-parameters, we employ our proposed module D_{out} in the output space of $LSD + D_{out}$. Due to the limited memory, the maximal resolution used in this experiment is of 256×128 . Thus we only conduct experiments on two kinds of sizes, i.e., 128×64 , 256×128 . Experimental results are shown in Table 5. According to the data, it can be observed that the module D_{out} has a positive effect on the final performance and can be generalized to other outstanding segmentation networks. The experimental results suggest that adapting the model in the output space performs better. In addition, it is interesting that the mIoU gain of 256×128 has a larger improvement than that of 128×64 . With the resolution of images increasing, there is no doubt that models could output more accurate predictions that contain abundant structure and location information. D_{out} is regarded as an auxiliary classifier based on the predictions $C(E(x^s))$ of source domain. As the resolution of images increases, D_{out} learns more precise knowledge from $C(E(x^s))$ and has a stronger ability to distinguish source predictions $C(E(x^s))$ from the target ones $C(E(x^t))$. Benefitting from the nice performance of D_{out} , the backbone updates parameters to force the data distribution of target domain close to source domain. This experiment showcases the effectiveness and generalization of the module D_{out} .

V. CONCLUSION

In this paper, we propose a new framework to address the challenging task of semantic segmentation for urban scenes. The proposed network primarily consists of three modules: the backbone, the image-based G-D pair and the output-based multi-category discriminator. The spotlight of this work is that unlike most of previous works, we do not impose discriminators on feature space for global alignment. Instead, the image-based G-D pair and the output-based multi-category discriminator are utilized to ensure that the proposed model has the ability to acquire high accurate labels from the learned feature representations. Ablation study suggests that it is a promising method to introduce the output-based multi-category discriminator in the output space of segmentation networks. In addition, this module can be combined with other outstanding segmentation networks for urban scenes to achieve higher performance. We evaluate the

proposed method on Cityscapes, using GTAV and STYNHIA as the source domain. The experimental results show that compared with some existing works, our method can achieve competitive performance conditioned on the same image size (i.e. 512×256), demonstrating the effectiveness of our proposed CAA-Net.

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